

MACHINE SHOP SECTOR 1 - LATHE L-1 CONDITION REPORT

Report No: MS1-2026-031 Date: 18 Aug 2026 Prepared by: V. Patil (shop engineer)

Machine: L-1 conventional centre lathe, Sahyadri SMT-450, commissioned 2009.

Swing over bed 450 mm, 1500 mm between centres, spindle motor 7.5 kW (efficiency class IE1).

1. CONDITION

Spindle runout is 0.045 mm at the chuck face against an acceptance limit of 0.02 mm.

Bed wear is 0.12 mm near the headstock; the tailstock is misaligned by 0.05 mm.

The chuck guard is fitted but is not interlocked with the spindle drive.

Headstock bearing noise was reported on 3 shifts in August.

2. RELIABILITY AND COST, LAST 12 MONTHS

Breakdowns: 6. Downtime: 142 hours. Repair cost: INR 3.9 lakh.

Like-for-like replacement cost (conventional lathe of the same capacity): INR 12.5 lakh.

3. UTILISATION

Utilisation was 78% of available hours over two shifts; 41 jobs were waiting for L-1.

4. SHOP ENGINEER'S NOTE

Accuracy is no longer adequate for pump shaft sleeves, which need 0.02 mm.

Recommend replacement with a CNC turning centre; requisition PR-2026-0418 raised.